

car) consists of 10 unique sensors with smart human-machine interface system associated with an android app, in simple words, we can get all the details of the vehicle on your mobile phone". The details displayed on the Android app include battery backup details, the temperature of the motor, axis details, accident alert, speed alert, fire and other safety systems.

Lastly, Patel invented AgriTech, a smart water sprinkler system for irrigation that can provide crops with a balanced ratio of water and manure using its AI and ML capabilities. It combats the issue of over-watering crops which leads to inefficient growth of the crops and waste of water.

### Nemkumar Bhantia

"India produces a bulk of tires every year which adds to the existing pollution issue when they are burned or incorrectly disposed of. We recycle over 300 million tires produced in the



Self-repairing road in Thondebavi village, Karnataka

country annually," stated Nemkumar Bhantia, Professor of Civil Engineering at the University of British Columbia. The Indian-origin innovator further adds that, along with his team, they recycle scrap tire fibre and mix it up with concrete to create roads in the rural areas of India.

These roads are also self-repairing and sustainable, meaning they have the ability to repair cracks that may develop over time automatically. Nemkumar and his team have worked extensively in Thondebavi village, 90 kilometres north of Bangalore, since 2014, testing and perfecting this technology. They successfully replaced a dilapidated dirt track which connected the village to the highway with the self-repairing road. The road

is about 100m in thickness which is 40 per cent less than that of a typical Indian road, which further diminishes costs and materials required. A large percentage of the cement is replaced with fly-ash thereby curbing the usual carbon footprint seen in typical Indian roads and the emission of greenhouse gases during the production of cement.

Explaining the road's built-in crack healing, Professor Nemkumar stated, "The fibres used in our roads have a hydrophilic nano-coating on them. This means they can attract or absorb water which in turn becomes available for crack healing. Usually, if typical roads develop a crack, they have unhydrated cement. However, our innovation gives the cement-fibre mixture the hydration capability which produces silicates that aid in closing the crack over time". Nemkumar's fibre-infused self-repairing roads cost 30 per cent less to make and can last for up to 15 years.

### Dr Govinder Singh Pawar

A scientist at the University of Exeter, UK, Dr Govinder Singh Pawar is leading the research to make fuel available from the process of photosynthesis. Pawar has fabricated

a semiconductor photoelectrode that can extract pure hydrogen from water by achieving photoelectrolysis. This form of hydrogen can be used as an alternative source of energy and fuel.

### Somil Daga

Putting his engineering studies to use, Somil Daga helps provide lighting solutions and affordable charging provisions to Gangapur village in Bihar. Along with his team, he attempts to address the energy needs of the village which, until Daga came along, had no electricity supply.

He has provided one LED per household in the village and designed a solar panel which can charge twenty mobile phones at the same time.

### Chewang Norphel

Chewang Norphel is an Indian civil engineer from Ladakh who has constructed 15 artificial glaciers hence earning the nickname 'Ice Man'. He created the glaciers by redirecting a river into a valley and slowing down the water stream by constructed obstacles. Once the flow has slowed down enough, the water freezes, creating man-made glaciers. These can be used to increase ground-water recharge and provide water for farming.

## ECO-CONSCIOUS INDIAN STARTUPS

These eco-conscious Indian startups are making an effort to overcome the environmental issues that torment our planet by introducing sustainable and eco-friendly solutions.

### K. K. Waste Management

"Plastic is inevitable. The banning of all plastic products is next to impossible because it is omnipresent. So, the best solution is to dispose of plastic in an efficient manner", says Mr Ahmed Khan, Managing Director at K.K. Plastic Waste Management and recipient of Real Hero Award, 2010.

Khan, along with his team, has come up with an innovative solution to dispose of plastic by blending it with synthetic vitamins or tar coal. This concoction is then put to use by K. K. Plastic Waste Management into asphaltting roads and pavements as well as making bricks and slabs. Khan and this team conceived this



Redirecting plastic waste into asphaltting roads.